

Two Lower-Sideband Phase Incompatibility

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Abstract

We isolate the first two orders of the punctured lower-even ladder, $m_j = k - j$ for $j = 2, 3$. At each order the exact direct coefficient is

$$p_j = Y_j + \mathcal{P}_j - S_j, \quad Y_j = \mathcal{T}_{k,m_j}^{\text{rest}} - d_{\sigma,k,2m_j}.$$

For fixed j ,

$$S_j = \frac{2m_j}{C_M} \beta^{2m_j} \{1 + o(1)\}, \quad \frac{\mathcal{P}_j}{S_j} \longrightarrow \gamma_j(\eta) = C_* C_M \lambda^{\eta-j}.$$

Thus $\gamma_3 = \gamma_2/\lambda$. Under the two explicit, presently unproved actual hypotheses $Y_j = o(H_{m_j})$, where $H_m = mR^{-2m}$, put $W_j = |p_j|/(2H_{m_j})$ and $x = (\beta R)^2 > 1$. If $a = \gamma_2(\eta) > 0$, then

$$\frac{W_2 + W_3}{x^{k-3}} \longrightarrow \frac{x|a-1| + |a/\lambda-1|}{C_M}.$$

The limit is uniformly positive. In fact,

$$\inf_{a>0} \max\{|a-1|, |a/\lambda-1|\} = \frac{\lambda-1}{\lambda+1},$$

whereas the sharper physical weighting gives

$$\inf_{a>0} \{x|a-1| + |a/\lambda-1|\} = 1 - \lambda^{-1}.$$

The second minimum occurs at $a = 1$. Hence the two-order direct subprefix diverges exponentially conditional on both named remainder hypotheses. Neither hypothesis, unconditional full-prefix nonclosure, growing-depth uniformity, nor any Gate A–E condition is proved. No Riemann-hypothesis conclusion is made.

1 The fixed two-sideband ledger

Retain the physical noise clock and phase

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \eta_\sigma = k - \frac{\log(1/\sigma)}{2 \log \lambda} \longrightarrow \eta. \quad (1)$$

For the two fixed labels $j \in \{2, 3\}$, set

$$m_j = k - j, \quad n_j = 2m_j, \quad H_m = mR^{-2m}, \quad x = (\beta R)^2 > 1. \quad (2)$$

The orders $2k - 4$ and $2k - 6$ lie below the two selected orders $2k$ and $2k - 2$, so both belong to the punctured lower-even family of RH-348 [4]. Abbreviate

$$p_j := p_{\sigma,k,2m_j}, \quad \mathcal{P}_j := \mathcal{P}_{\sigma,2m_j}. \quad (3)$$

The simultaneous complete-orbit extraction and the deterministic numerator anchor give the exact identities

$$p_j = Y_j + \mathcal{P}_j - S_j, \quad Y_j = \mathcal{T}_{k,m_j}^{\text{rest}} - d_{\sigma,k,2m_j}, \quad S_j = F_{m_j}^{\text{orb}} + \mathcal{A}_{k,2m_j}. \quad (4)$$

Here

$$F_m^{\text{orb}} = 2mG_m, \quad G_m = \frac{r_H^{-2m}}{1 + |M_m|}, \quad (5)$$

$$\mathcal{A}_{k,2m} = 2(\beta^{2m} - \beta_k^{2m}), \quad \beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H\sqrt{\lambda}}. \quad (6)$$

The p_j are direct coefficients. In particular, the noisy-head/counterloop defect remains inside the actual signed quantity Y_j ; it is not silently discarded or reclassified as a full-trace term [3, 5].

Proposition 1.1 (Fixed-sideband demand and parity laws). *For each fixed $j \in \{2, 3\}$,*

$$S_j = \frac{2m_j}{C_M} \beta^{2m_j} \{1 + o(1)\}, \quad (7)$$

$$\frac{S_j}{2H_{m_j}} = \frac{x^{m_j}}{C_M} \{1 + o(1)\}, \quad (8)$$

$$\frac{\mathcal{P}_j}{S_j} \longrightarrow \gamma_j(\eta) := C_* C_M \lambda^{\eta-j}. \quad (9)$$

Consequently,

$$\boxed{\gamma_3(\eta) = \lambda^{-1} \gamma_2(\eta)}. \quad (10)$$

Proof. The boundary multiplier law is $M_m = -C_M \lambda^m \{1 + o(1)\}$, so

$$G_m = C_M^{-1} \beta^{2m} \{1 + o(1)\}, \quad F_m^{\text{orb}} = \frac{2m}{C_M} \beta^{2m} \{1 + o(1)\}.$$

Also

$$\beta_k = \beta \exp \left[-\frac{\log C_M}{2k} + o(k^{-1}) \right].$$

Since $m_j/k \rightarrow 1$ for fixed j , equation (6) gives $\mathcal{A}_{k,2m_j} = O(\beta^{2m_j})$. This is $O(m_j^{-1})$ relative to $F_{m_j}^{\text{orb}}$, proving (7). Division by $2H_{m_j} = 2m_j R^{-2m_j}$ proves (8) [1, 2, 7].

For even order $2m$, the exact parity packet is

$$\mathcal{P}_{\sigma,2m} = r_H^{-2m} \{1 - (1 - \delta_\sigma)^{2m}\}, \quad \delta_\sigma = C_* \sqrt{\sigma} + o(\sqrt{\sigma}). \quad (11)$$

The uniform quadratic remainder in RH-326 and $m_j \sqrt{\sigma} \rightarrow 0$ yield

$$\mathcal{P}_j = 2m_j C_* \sqrt{\sigma} r_H^{-2m_j} \{1 + o(1)\}.$$

After division by (7), the remaining clock factor is

$$\sqrt{\sigma} \lambda^{m_j} = \lambda^{\eta_\sigma - k + m_j} = \lambda^{\eta_\sigma - j}.$$

This proves (9); (10) is immediate [3, 6]. □

Every limit in proposition 1.1 is pointwise in the fixed label j . No uniformity for a depth $j = j_k \rightarrow \infty$ is asserted.

2 Two exact phase minimax laws

Write

$$a = \gamma_2(\eta) = C_* C_M \lambda^{\eta-2} > 0. \quad (12)$$

Then $\gamma_3 = a/\lambda$. The first minimax law measures the larger relative mismatch at the two coordinates.

Proposition 2.1 (Relative two-sideband minimax). *For every $\lambda > 1$,*

$$\boxed{\inf_{a>0} \max\{|a-1|, |a/\lambda-1|\} = \frac{\lambda-1}{\lambda+1}.} \quad (13)$$

The unique minimizer is $a_{\text{rel}} = 2\lambda/(\lambda+1)$.

Proof. For $0 < a \leq 1$, the larger term is $1 - a/\lambda$, which decreases to $1 - 1/\lambda$. For $1 \leq a \leq \lambda$, the terms are respectively $a - 1$ and $1 - a/\lambda$. Their maximum is minimized at their unique intersection,

$$a - 1 = 1 - a/\lambda, \quad a = \frac{2\lambda}{\lambda+1},$$

where both equal $(\lambda-1)/(\lambda+1)$. For $a \geq \lambda$, the maximum is $a - 1$ and increases. The intersection value is strictly smaller than the boundary values, proving the claim. $\square$

The prefix normalization does not weight these coordinates equally: $x^{m_2}/x^{k-3} = x$, whereas $x^{m_3}/x^{k-3} = 1$.

Proposition 2.2 (Physical prefix-weighted minimax). *If $\lambda > 1$ and $x > 1$, then*

$$\boxed{\inf_{a>0} \{x|a-1| + |a/\lambda-1|\} = 1 - \lambda^{-1}.} \quad (14)$$

The unique minimizer is $a_{\text{wt}} = 1$.

Proof. For $0 < a \leq 1$, the objective is

$$x + 1 - a(x + \lambda^{-1}),$$

which is strictly decreasing. For $1 \leq a \leq \lambda$, it is

$$x(a-1) + 1 - a/\lambda,$$

whose slope $x - \lambda^{-1}$ is positive. For $a \geq \lambda$, it is $x(a-1) + a/\lambda - 1$, again strictly increasing. Hence the unique global minimum occurs at $a = 1$ and equals $1 - \lambda^{-1}$. $\square$

The two optimizers differ. Equalizing the relative mismatches is not the same operation as minimizing the actual prefix-weighted sum.

3 Conditional physical two-order obstruction

Define the two actual weighted direct contributions

$$W_j = \frac{|p_j|}{2H_{m_j}} = \frac{|p_j|R^{2m_j}}{2m_j}, \quad j = 2, 3. \quad (15)$$

Theorem 3.1 (Conditional two-lower-sideband divergence). *Suppose an actual physical sequence satisfies (1) and, simultaneously,*

$$\boxed{Y_2 = o(H_{m_2}), \quad Y_3 = o(H_{m_3}).} \quad (16)$$

Then for each $j = 2, 3$,

$$\frac{W_j}{x^{m_j}} \longrightarrow \frac{|\gamma_j(\eta) - 1|}{C_M}. \quad (17)$$

Consequently, with a as in (12),

$$\boxed{\frac{W_2 + W_3}{x^{k-3}} \longrightarrow \frac{x|a-1| + |a/\lambda-1|}{C_M} \geq \frac{1 - \lambda^{-1}}{C_M} > 0.} \quad (18)$$

In particular, $W_2 + W_3$ diverges exponentially.

Proof. Equation (8) and $x > 1$ imply $H_{m_j}/S_j \rightarrow 0$. Therefore each hypothesis in (16) gives $Y_j/S_j \rightarrow 0$. Divide (4) by S_j and apply (9):

$$\frac{p_j}{S_j} \rightarrow \gamma_j(\eta) - 1.$$

Multiplication by (8) proves (17), including the case in which the limiting mismatch is zero. Since $m_2 = k - 2$, $m_3 = k - 3$, and $\gamma_3 = a/\lambda$,

$$\frac{W_2 + W_3}{x^{k-3}} = x \frac{W_2}{x^{m_2}} + \frac{W_3}{x^{m_3}}.$$

Taking limits and applying proposition 2.2 proves (18). $\square$

Corollary 3.2 (Bounded-phase uniform lower law). *Suppose only that η_σ remains in a fixed bounded interval, and assume the same two simultaneous actual remainder hypotheses (16). Then*

$$\boxed{\liminf_{\sigma \rightarrow 0} \frac{W_2 + W_3}{x^{k-3}} \geq \frac{1 - \lambda^{-1}}{C_M} > 0.} \quad (19)$$

Proof. On a bounded phase interval, the quadratic parity remainder used in proposition 1.1 is uniform for each of the two fixed labels. Put $a_\sigma = C_* C_M \lambda^{\eta_\sigma - 2}$. The proof of theorem 3.1, without taking a phase limit, gives

$$\frac{W_2 + W_3}{x^{k-3}} = \frac{x|a_\sigma - 1| + |a_\sigma/\lambda - 1|}{C_M} + o(1).$$

Apply proposition 2.2 pointwise in a_σ and take the lower limit. $\square$

Remark 3.3 (The hypotheses are not conclusions). The quantities Y_j contain the actual signed orbit-free trace remainder and noisy-head/counterloop defect at their respective moving orders. The repository proves neither estimate in (16). Thus theorem 3.1 is a conditional theorem about two actual direct coefficients, not an unconditional physical nonclosure result. It does not provide an unconditional verdict for the full $\mathcal{E}_{\text{off}}$ aggregate.

4 Executable audit and route boundary

The artifact reconstructs the two boundary multipliers, complete orbit atoms, radial terms, demands, and exact parity-formula evaluations under the leading-law scalar fixture at $k = 10, 18, 30, 46$. It chooses the physically weighted optimizer $a = 1$, uses $Y_2 = Y_3 = 0$, and verifies the approach of

$$C_M \frac{W_2 + W_3}{x^{k-3}} \quad \text{to} \quad 1 - \lambda^{-1}.$$

It also checks both exact minimax formulas and their distinct optimizers. These decimal rows reproduce formulas only. They are not interval certificates, asymptotic evidence, measurements of a noisy operator, or evidence for (16).

RH-350 may audit a fixed-depth extension $j = 2, \dots, J$ and, separately, ask whether any slowly growing $J = J_k$ is supported. The latter requires new uniform control of the parity Taylor remainder, multiplier law, radial term, ratio m_j/k , and simultaneous actual remainder hypotheses. None is inferred from this fixed two-order theorem.

Odd orders, upper off-alias orders, actual signed remainder control, and unconditional full-direct-prefix behavior remain open, as do determinant gluing and RH-288. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

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